

Predicting Molecular Reality in 4D

The first World Model + LLM fusion platform for 4D dynamic drug discovery.

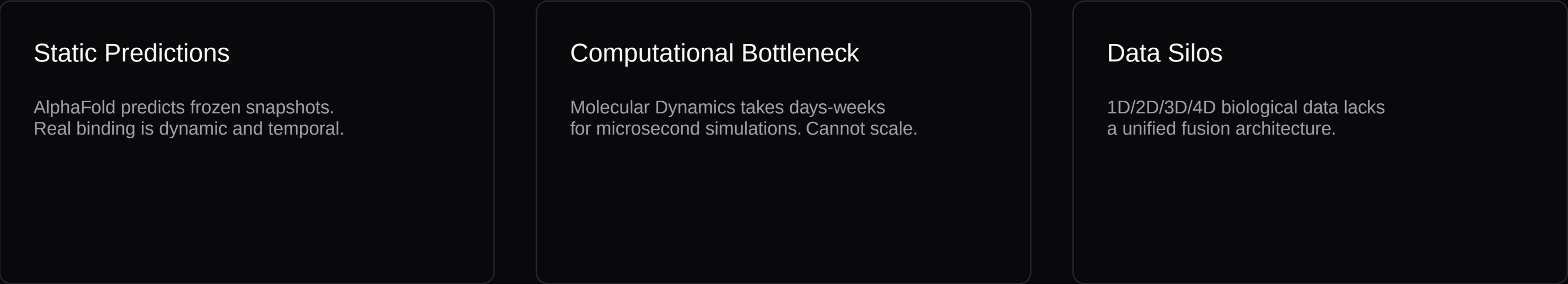
voxeltox.space

Seed Round | \$3M–\$5M | 2026

THE PROBLEM

Drug Discovery Is Broken

Current AI-driven drug discovery relies on static predictions that cannot capture the dynamic reality of molecular interactions.



\$2.6B

avg cost per drug

>90%

clinical trial failure rate

\$70B+

market size by 2030

OUR SOLUTION

VoxelTox Platform

World Model + LLM fusion for 4D dynamic deterministic prediction.
10,000x faster than traditional molecular dynamics.

4D Occupancy Forecasting

Predict WHERE atoms will be over time — not HOW they got there. Replaces trajectory integration.

Bi-directional Self-Correction

LLM proposes candidates, World Model validates in virtual physics environment. Continuous refinement loop.

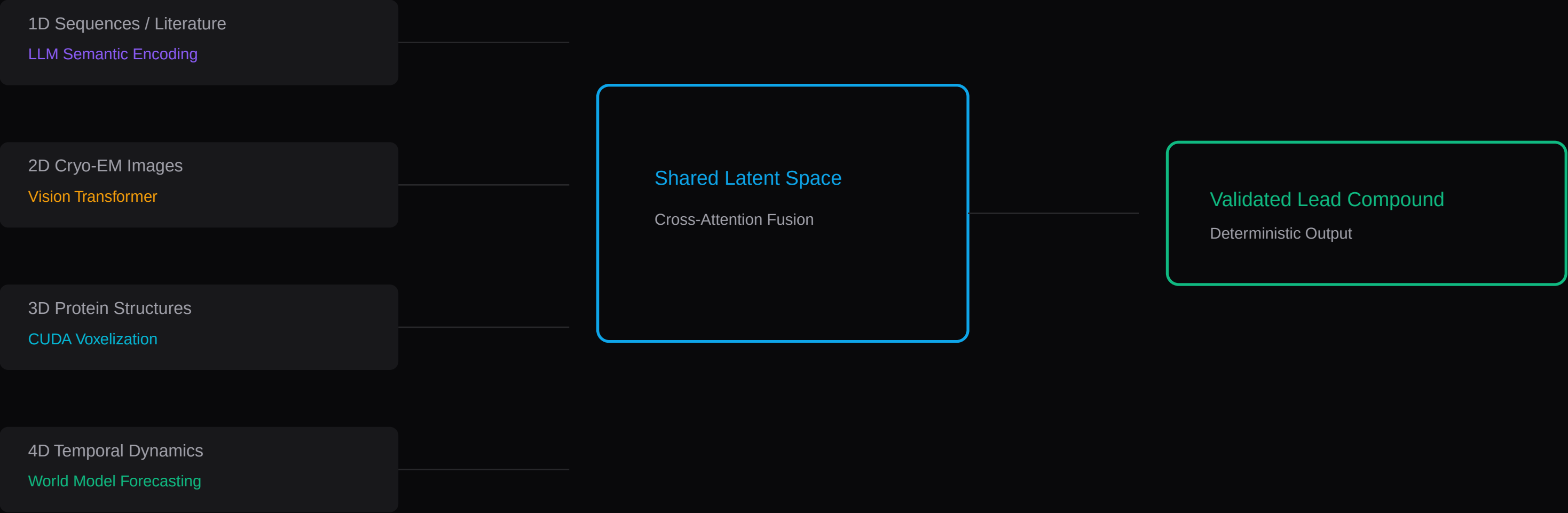
Native Multi-Modal Fusion

Fuses 1D sequences, 2D Cryo-EM, 3D structures, and 4D dynamics in shared latent space.

Physics-Informed ML

Learns molecular behavior from data instead of solving quantum equations. Orders of magnitude faster.

Multi-Modal Fusion Architecture



GPU-Accelerated Computing

Designed from the ground up for massively parallel GPU execution.

Training	Multi-GPU distributed (H100), mixed-precision BF16, FSDP/DeepSpeed
Inference	TensorRT-optimized, sub-second 4D occupancy prediction
Data Pipeline	Custom CUDA kernels, RAPIDS-accelerated preprocessing
Frameworks	PyTorch, CUDA-X (cuBLAS, cuFFT, cuDNN), Triton inference serving
Pretrained Models	Fine-tuned from Llama/Mistral (LLM), ViT/DINOv2 (Vision)

Estimated Total Compute:

35,000 – 50,000 H100 GPU-Hours

over 24 months

Clearly Defined Datasets

3D Protein Data Bank (PDB)

220K structures

3D AlphaFold DB

200M predicted structures

2D EMDB

40K Cryo-EM maps

1D ChEMBL

2.4M+ bioactive compounds

1D PubMed / MEDLINE

36M+ research papers

1D ZINC Database

230M+ compounds

4D MoDEL / mdCATH

Curated MD trajectories

2D EMPIAR

Raw EM image archives

Project Milestones

Phase 1 Month 1–6	Foundation Data pipeline, World Model v1, LLM fine-tuning	PoC on known protein-ligand pairs ~12K GPU-hrs
Phase 2 Month 7–12	Integration ViT adaptation, fusion layer, bi-directional loop	End-to-end pipeline on benchmarks ~15K GPU-hrs
Phase 3 Month 13–18	Validation Large-scale training, inference optimization	Publication-ready results, first pilot ~15K GPU-hrs
Phase 4 Month 19–24	Production SaaS deployment, customer engagements	Revenue-generating platform ~8K GPU-hrs

Path to Revenue

Year 1–2: Compute SaaS

High-precision 4D dynamic virtual screening as a service for top pharma and research institutions. Subscription + per-computation pricing.

Year 2–4: Joint Pipelines

Co-develop lead compounds with wet-lab partners. Shared IP, milestone payments, and downstream royalties.

Year 4+: Platform Licensing

License VoxelTox as the industry-standard spatial prediction engine. Platform licensing + API access fees.

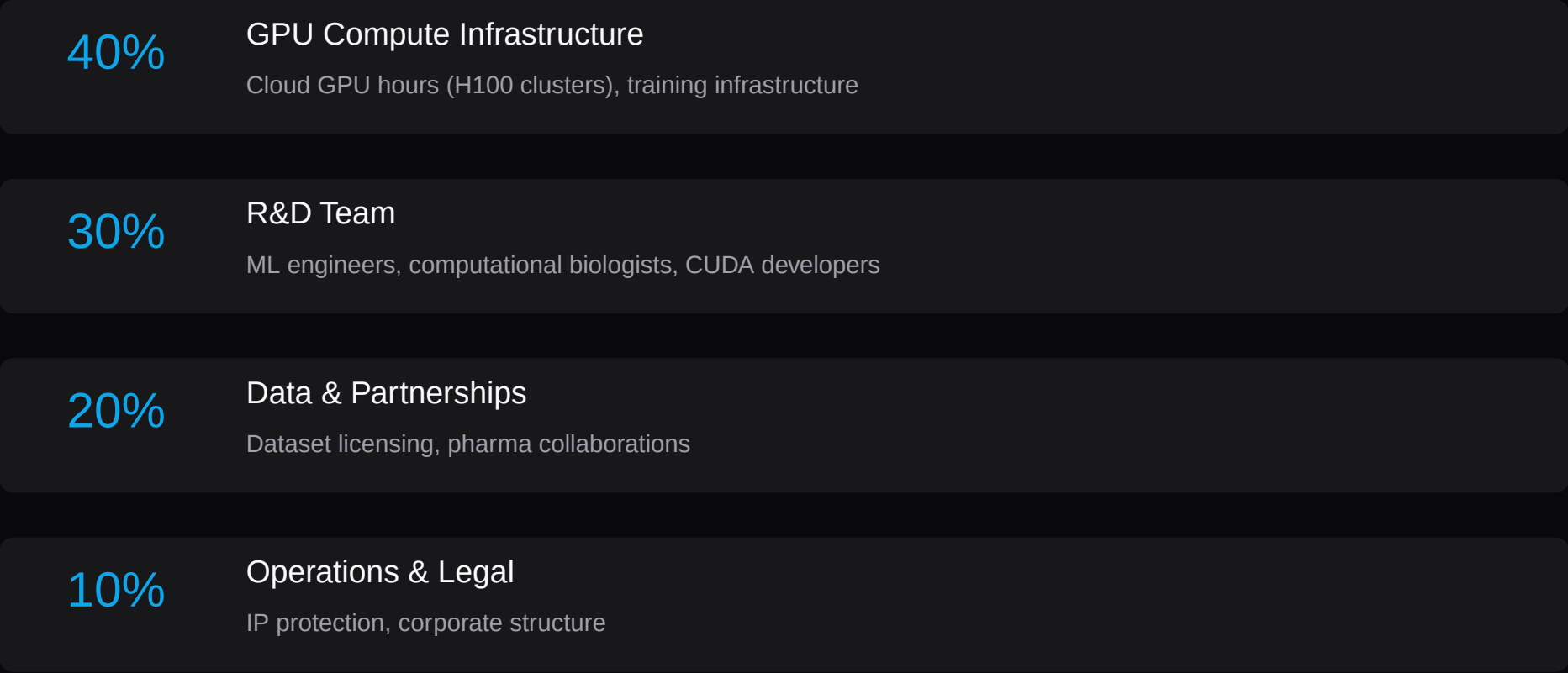
Competitive Landscape

Competitor	Approach	Key Limitation
AlphaFold / DeepMind	Static structure prediction	No dynamics, no binding kinetics
Schrodinger	Physics-based MD	Too slow to scale
Recursion Pharma	High-throughput wet lab	No spatial prediction
Insilico Medicine	Generative chemistry	No 4D physics validation
Relay Therapeutics	MD + ML hybrid	Still equation-dependent

VoxelTox Unique Position

Only platform combining LLM + World Model | Native 1D/2D/3D/4D fusion | GPU-native architecture | Open-source commitment

Seed Round: \$3M–\$5M



Beyond Capital:

- GPU/compute asset access
- Cloud ecosystem integration
- Pharma R&D introductions
- Scientific advisory connections

Social Impact & Open Source

- Accelerating drug discovery reduces time-to-market for life-saving treatments
- Reducing clinical trial failure rates lowers healthcare costs globally
- Open-source CUDA kernels advance scientific computing for all researchers
- Democratizing 4D molecular prediction for academic labs worldwide
- Applicable to neglected tropical diseases and rare disease targets

Open-Source Commitment

- Core voxelization CUDA kernels
- Benchmark datasets for 4D molecular occupancy prediction
- Pretrained model checkpoints for academic research
- Papers targeting NeurIPS, ICML, ICLR

Team & Contact

Tensor

Founder & CEO

Multi-modal sensor fusion & spatial computing
World Models, LLM systems, CUDA development
Autonomous driving perception !' molecular space

Actively hiring:

- ML Engineers
- Computational Biologists
- CUDA / Systems Architects

Contact

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